Maiqi Jiang

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Focus: What interventions actually change inside an LLM (structured pruning, knowledge editing, adapters, decoding), and how to build interventions that preserve the capabilities that matter. Mechanistic analysis, RL-based methods, and released tooling.

Education

- **College of William & Mary** — Williamsburg, VA
  PhD in Computer Science (Advisor: Prof. Yanfu Zhang) — *8.2025 - Present*
- **The Hong Kong Polytechnic University (QS Top 60)** — Hong Kong
  *M.Sc. in Information Technology with **Distinction Award*** — *8.2021 - 1.2023*
- **Northeastern University, China (985 Project)** — Shenyang, China
  B.S. in Biomedical Engineering — *9.2017 - 6.2021*

Publications

- **Maiqi Jiang**, Yanshuo Chen, Guodong Liu, Avinash Sahu, Ye Gao, Yanfu Zhang: PAIRS, Not Labels: Predicting Protein-Phenotype Associations via Link Prediction. IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP 2026**).
- Mahmud Wasif Nafee[*], **Maiqi Jiang**[*], Haipeng Chen, Yanfu Zhang: Dynamic Retriever for In-Context Knowledge Editing via Policy Optimization. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025**), pp. 16744–16757. ([*]Co-first Author)
- Xuzhong Wang, **Maiqi Jiang**, Tejal Nair, Girija Bhusal, Yanfu Zhang, Haipeng Chen: Towards Reliable, Generalizable, and Specific In-Context Knowledge Editing via Multi-Objective Reinforcement Learning. **EMNLP 2026 Findings**.
- Tejal Nair, Mahmud Wasif Nafee, **Maiqi Jiang**, Ashley Gao, Haipeng Chen, Yanfu Zhang: Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing. Findings of the Association for Computational Linguistics: **ACL 2026 Findings**.
- Omid Mersa, **Maiqi Jiang**, Ye Gao, Qun Li, Yanfu Zhang: Hierarchical Convolution Multibranch Transformer for EEG Signals. IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP 2026**).
- **Maiqi Jiang**, Xugui Zhou, Bin Ren, Guodong Liu, Yanfu Zhang: In-Context Task Learning: Reading Out a Kernel Classifier. Submitted to **ICLR 2027** (under review).
- **Maiqi Jiang**, Xugui Zhou, Bin Ren, Junyi Li, Shangqian Gao, Yanfu Zhang: SFT-P: Federated Tuning and Pruning for LLMs. Submitted to **NAACL 2027**.
- Yiran Ding, **Maiqi Jiang**, Qun Li, Yanfu Zhang: One Direction Away: The Cross-Lingual Gap in Long-Context Reasoning and a Shift That Closes It. Submitted to **NAACL 2027**.
- Tejal Nair, Omid Mersa, Zhen Xu, **Maiqi Jiang**, Xuzhong Wang, Ashley Gao, Gang Zhou, Haipeng Chen, Yanfu Zhang: Representation Change and Adaptive Stopping in In-Context Learning. Submitted to **NAACL 2027**.
- **Maiqi Jiang**, Noman Ali, Yiran Ding, Yanfu Zhang: Is Meta-Path Attention an Explanation? Evidence of Alignment and Decoupling in Heterogeneous GNNs. Submitted to **TMLR** (under review).
- Jiarui Li, **Maiqi Jiang**, Junyi Li, Bin Wang, Shangqian Gao, Yanfu Zhang: Disentangling Memorization in Language Models via Sparse Representation Intervention. Submitted to **Neural Networks** (under review).
- Yiran Ding, Muhammed Muminul Hoque, **Maiqi Jiang**, Yongming Qin, Sidi Lu, Yanfu Zhang: DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction. Submitted to **ICRA 2027** (under review).
- Zhaoqing Li[*], **Maiqi Jiang**[*], Shengyuan Chen, Bo Li, Guorong Chen, Xiao Huang: Automated Heterogeneous Network Learning with Non-Recursive Message Passing. arXiv preprint, 2025. ([*]Co-first Author)

Research Experience

- **Federated Tuning & Structural Pruning for LLMs (SFT-P)** — William & Mary
  Researcher (Advisor: Prof. Yanfu Zhang) — *Aug 2025 – Jan 2026*
  - Proposed SFT-P: **FedAvg**-based training that jointly learns client-specific structured pruning masks and adaptation for **on-device** deployment under non-IID private data; supports mixed client budgets.
  - Implemented a **client-conditioned mask generator** (shared hypernetwork + private client embedding) producing hard binary routing masks via Bernoulli gating, and a residual-safe route/merge interface for dense-kernel execution; achieved **+8.5 Avg** over the best federated baseline on **LLaMA-7B at 50% pruning**.
- **In-Context Knowledge Editing for LLMs (DR-IKE / FWE-IKE / MO-IKE)** — William & Mary
  Researcher (Advisors: Prof. Yanfu Zhang, Prof. Haipeng Chen) — *Aug 2025 – Present*
  - Main contributor to a research line on **retrieval-based in-context knowledge editing** for black-box LLMs, spanning policy-optimized retrieval, confidence-aware fusion, and constrained multi-objective RL.
  - **DR-IKE** (EMNLP 2025, co-first): trained a BERT retriever via REINFORCE with a **learnable threshold** to dynamically size demonstrations. [code]
  - **FWE-IKE** (ACL 2026 Findings): **confidence-gated ensemble** of LLM/BERT/MLP rankers with logit-space fusion, closing ∼53% of the gap to the oracle on COUNTERFACT.
  - **MO-IKE** (EMNLP 2026 Findings): formulated prompt construction as a **Constrained MDP** with Lagrangian relaxation and GRPO, jointly optimizing reliability, generality, and specificity.
- **Meta-Path Attention Reliability in Heterogeneous GNNs (MetaXplain)** — William & Mary
  Researcher (Advisor: Prof. Yanfu Zhang) — *Aug 2025 – Feb 2026*
  - Conducted a controlled empirical study of whether **meta-path attention is a faithful explanation**, identifying regimes of **alignment vs. statistically significant decoupling** across multiple datasets and HGNN backbones.

- Introduced **MetaXplain**, a meta-path-aware post-hoc analysis protocol (view-factorized explanations, schema-valid perturbations, fusion-aware attribution), and proposed **MP-AEA** to quantify attention–explanation alignment; released an explainability benchmark codebase (ExplainerForHGNN) for the community. [ExplainerForHGNN]

- **In-Context Task Learning: Reading Out a Kernel Classifier** — William & Mary
  Project Lead (Advisor: Prof. Yanfu Zhang) — *Feb 2026 – Present*
  - Identified **mapping-sensitive attention heads** through paired intact/randomized-label contrasts and characterized their outputs as a **kernel classifier** combining query-dependent retrieval with contextual class votes. On **TREC-fine** with Llama-3.1-8B and 180 demonstrations, direct readout from **8 selected heads** achieves **50.3% accuracy**, versus 51.6% for the full model.
  - Proposed **Selective TL Memory**, reusing cached demonstration K/V through selected heads with live query-dependent retrieval. Selective access improves the same fixed readout across **all 18 natural-language settings** at both tested head budgets; the **30-head readout** outperforms all compared adapted vector baselines in **14/18 settings**. Submitted to **ICLR 2027** (under review).

- **DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction** — William & Mary
  Project Mentor (Advisor: Prof. Yanfu Zhang) — *Dec 2025 – Mar 2026*
  - Mentored two interns (the first two authors); guided the design of **DyV2X**, a cooperative trajectory prediction framework that constructs **per-timestep** dynamic interaction graphs and uses a **temporal graph attention encoder** to jointly model evolving inter-agent interactions and individual kinematic history.
  - On V2X-Seq-TFD (9,320 target agents), improved over the V2X-Graph baseline on all metrics: **ADE** 1.110→0.952 (**14.2%**↓), **FDE** 1.894→1.664 (**12.1%**↓), **MR@2m** 27.9%→24.7%.
  - Largest gains on safety-critical **turning maneuvers** (e.g., *left turn*: FDE 4.216→2.646, **37.2%**↓; MR@2m 64.2%→47.5%), where inter-agent interactions evolve most rapidly.
  - Submitted to **ICRA 2027** (under review).

- **Mentorship: Intern-led Projects on LLM Efficiency & Long-Context** — William & Mary
  Project Mentor (Advisor: Prof. Yanfu Zhang) — *Sep 2025 – Present*
  - **Domain-Adaptive Pruning (CDAM).** Mentoring *Noman Ali* on a **training-free activation measure** combining domain-specific and general-purpose statistics with feature-wise cross-domain interactions, integrated into Wanda and CFSP. On Llama-2-13B at **50% sparsity** with 1,024 calibration examples, CDAM raises CFSP's average biomedical accuracy from **34.45% to 38.70%** versus naive calibration mixing, while improving general-task accuracy from 58.73% to 61.32%. Targeting **ACL 2027**.
  - **ICL-Preserving Pruning (ICLPrune).** Mentoring *Chongyuan Sun* on improving pruned-LLM **in-context learning** performance under task-shifted prompts via demonstration-aware calibration (*ICLPrune: Improving In-Context Learning in Structured LLM Pruning via Demonstration-Aware Calibration*, targeting **ACL 2027**).
  - **Long-Context & ICL.** Proposed the research direction and co-supervised (with Prof. Zhang) *Yiran Ding* on the **cross-lingual gap in long-context reasoning** under context-extension methods, leading to *One Direction Away: The Cross-Lingual Gap in Long-Context Reasoning and a Shift That Closes It* (submitted to **NAACL 2027**); mentoring *Muhammed Muminul Hoque* on demonstration-ordering policies for ICL motivated by the **Lost-in-the-Middle** effect, given per-example rankings.

- **lmdiff: Behavioral Diagnostics for LLM Configurations** — Independent / Open Source
  Project Lead (Advisor: Prof. Yanfu Zhang) — *Oct 2025 – Present*
  - Built and released **lmdiff** (PyPI `lmdiff-kit`, GitHub), a Python library for comparing LLM **configurations**—weights + system prompt + decoding + adapter—via behavioral distance and multi-level diagnostics, rather than aggregate benchmark scalars.
  - Centered the API on a `family()` abstraction that compares one base configuration against N variants on a shared set of probe-domain axes, treating weight modifications (fine-tuning, RLHF, adapters), **runtime modifications** (system prompt, decoding, ICL context), and quantization/pruning as first-class variants. Geometry-level metrics (per-domain drift, share, raw + selective cross-variant cosine) quantify *where* each variant moves and whether different modifications push in the same direction—a view a pairwise BD scalar or single-model benchmark score cannot produce. Showcase 7-variant Llama-2-7B family reveals a system prompt can reshape next-token distributions more than 7B→13B scaling.
  - Engineering: **engine layer** with HF Transformers + mock backends, **engine reuse** that keeps peak VRAM at base + 1 active variant in 7-variant families, **lm-eval-harness** multi-task probe loader with task→domain mapping, JSON schema versioning with auto-recompute on load, 5-channel reporting (ANSI / PNG / HTML / Markdown / JSON), ∼860 tests. Active development; representation, trajectory, and causal metrics planned for v0.4+.

- **Automated Heterogeneous Network Learning (AutoGNR)** — The Hong Kong Polytechnic University
  Researcher (Advisor: Prof. Xiao Huang) — *Nov 2021 – Feb 2023*
  - Co-developed **AutoGNR**, using **differentiable NAS** to automatically select heterogeneous aggregation paths and node-type combinations—conceptually aligned with **structured pruning** and bi-level optimization. [AutoGNR] [HINBaselines]
  - Designed a non-recursive message-passing scheme that decouples path search from depth, improving search efficiency on large heterogeneous graphs.

- **Temporal User Engagement Modeling in Polarized Social Platforms** — City University of Hong Kong
  Research Assistant (Advisor: Prof. Junming Liu) — *Feb 2023 – Dec 2023*
  - Fused **image/video content features** with text and interaction context via a **temporal GNN** for forecasting user engagement across polarized communities on Twitter/Reddit.

- Built multimodal feature pipelines and longitudinal datasets for studying engagement dynamics under polarization. [RedditProcess] [RedditDataset]
- Submitted as a short paper to **ICIS 2023**, proposing a counter-party graph attention network with sentiment diffusion for online engagement prediction.

- **Semi-supervised Learning for Whole-Slide Pathology Images (WSI)** Northeastern University (China)
  Undergraduate Researcher / Bachelor's Thesis (Advisor: Prof. Xiaoyu Cui) *Nov 2018 – Jun 2021*
  - Developed weakly-/semi-supervised pipelines for gigapixel WSI using eye-tracking signals, focusing on **efficient patch sampling** and **large-scale image processing**. [Modeling] [Processing]
  - Maintained reusable WSI tooling (tiling, feature extraction, dataset management) used across the lab's projects.

Skills

- **LLM Training & Adaptation**: Federated LLM tuning with FedAvg; structured pruning with learned binary masks and dense-kernel route/merge execution (LLaMA-7B); HuggingFace + PyTorch.
- **RL for LLMs**: REINFORCE-based and GRPO-based retrieval / policy optimization for in-context knowledge editing (DR-IKE, MO-IKE); reward design under Constrained MDP with Lagrangian relaxation.
- **Graph Learning**: Heterogeneous GNNs (HAN/HGT-family), temporal graph networks (TGN / DyGFormer), differentiable NAS over heterogeneous aggregation paths; PyG-style implementation.
- **Mechanistic Interpretability**: Attention head analysis for ICL; sparse autoencoder (SAE) representation interventions with ReFT; meta-path attention faithfulness diagnostics.
- **Systems & Tooling**: Python library design and PyPI release (`lmdiff-kit`); multi-codebase open-source benchmarks (ExplainerForHGNN, HINBaselines); large-scale data pipelines for WSI and multimodal social media; mentoring code reviews across 4 intern projects.

Industry & Service

- **Research Assistant**: City University of Hong Kong, with Prof. Junming Liu (Feb 2023 – Dec 2023)
- **Software Developer Intern**: Neusoft Inc., China (Sep 2020)
- **PC member/Reviewer**: ICASSP 2026, NeurIPS 2026, CIKM 2026 Workshop, ICLR 2027, ACM Transactions on Computing for Healthcare
- **Open Source Contributor**: The Unified AI Framework (Jul 2023); ExplainerForHGNN benchmark (2025–2026)